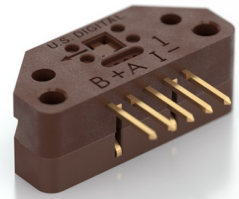




EM2 Features

- Encoder module for measuring rotary or linear movement
- Two-channel quadrature with index channel
- 11 Rotary Resolutions from 2,000 to 10,000 CPR (8,000 to 40,000 PPR)
- 2 Linear resolutions from 1,000 to 2,000 LPI (x1 quadrature decoding)
- Sink/source 5 mA
- Internal decoupling capacitor for enhanced noise immunity
- Single 5V supply



EM2 Product Description

The EM2 is a transmissive optical encoder module designed to measure rotary or linear movement when paired with an encoder disk or linear strip. The EM2 consists of a lensed LED source and a monolithic detector IC enclosed in a small polymer package. The EM2 uses phased array detector technology, allowing for a wider gap tolerance and reliable performance even with some disk contamination.



The EM2 encoder module provides digital A & B quadrature outputs with an index channel. Each EM2 module is resolution-specific and is matched to the resolution of an encoder disk or linear strip. The EM2 module supports 12 different resolutions. The EM2 operates with a single 5V supply and provides single-ended outputs capable of sinking and sourcing 5mA. An internal 0.1 μ F decoupling capacitor provides enhanced noise immunity.

For open collector and higher voltage applications, add the PC3 (<https://www.usdigital.com/products/accessories/interfaces/cable-drivers/pc3/>) cable driver, or for differential cable driver outputs, add the PC4 (<https://www.usdigital.com/products/accessories/interfaces/cable-drivers/pc4/>) cable driver.

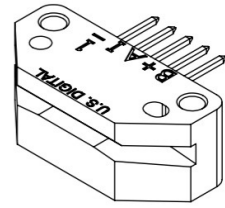
Accessories such as encoder disks, linear strips, mating connectors, and cables are also available.

Download (<https://www.usdigital.com/media/4tvlmrwp/em2-em1-comparison.pdf>) the EM2/EM1 comparison chart.

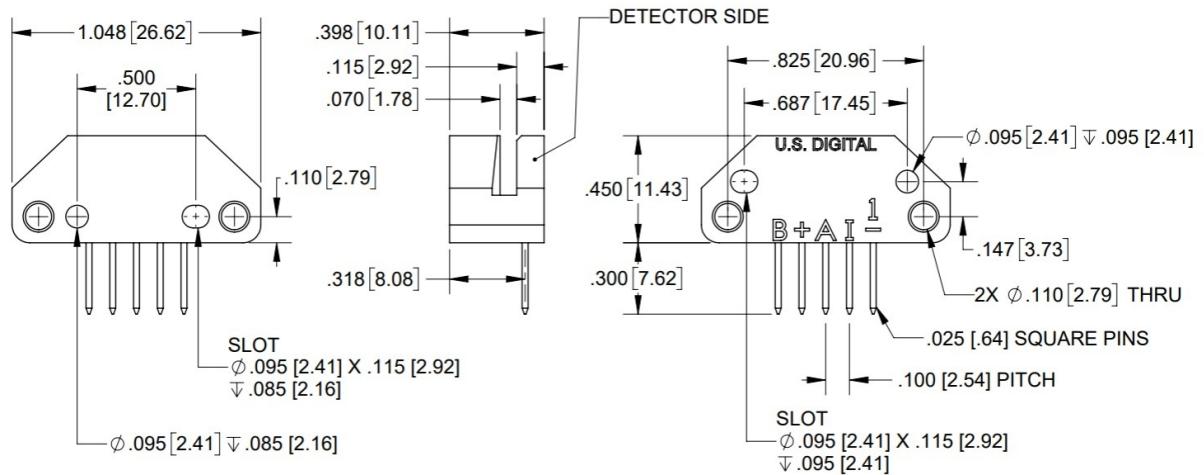
Mechanical Drawings

EM2 Transmissive Optical Encoder Module

EM2 Transmissive Optical Encoder Module



RELEASE DATE: 02/07/2022



1400 NE 136th Avenue
Vancouver, Washington 98684, USA

info@usdigital.com
www.usdigital.com

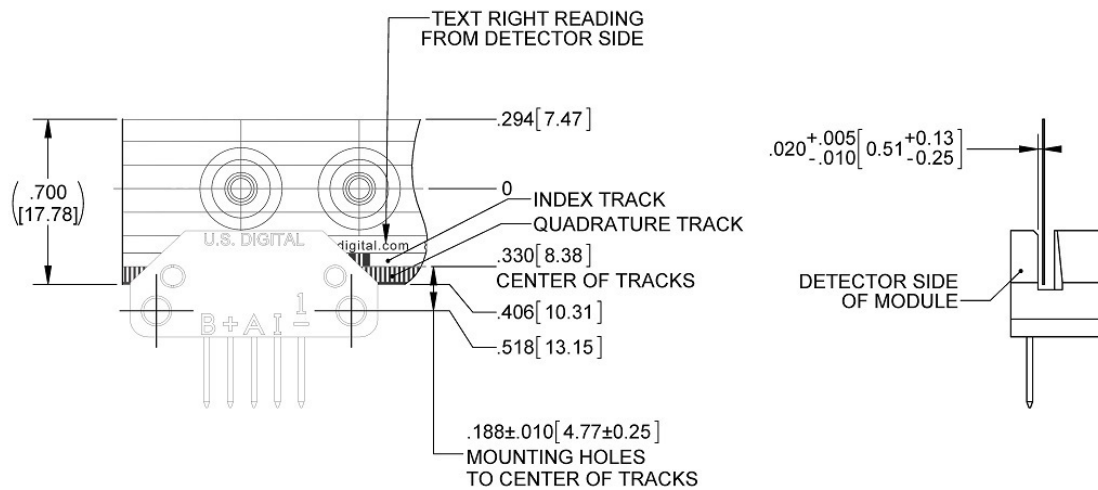
Local: 360.260.2468
Toll-free: 800.736.0194

UNITS: INCHES [MM]
METRIC SHOWN FOR REFERENCE ONLY

| EM2 Transmissive Optical Encoder Module

EM2 Transmissive Optical Encoder Module Linear Strip Alignment

RELEASE DATE: 04/07/2015



US DIGITAL 1400 NE 136th Avenue
Vancouver, Washington 98684, USA

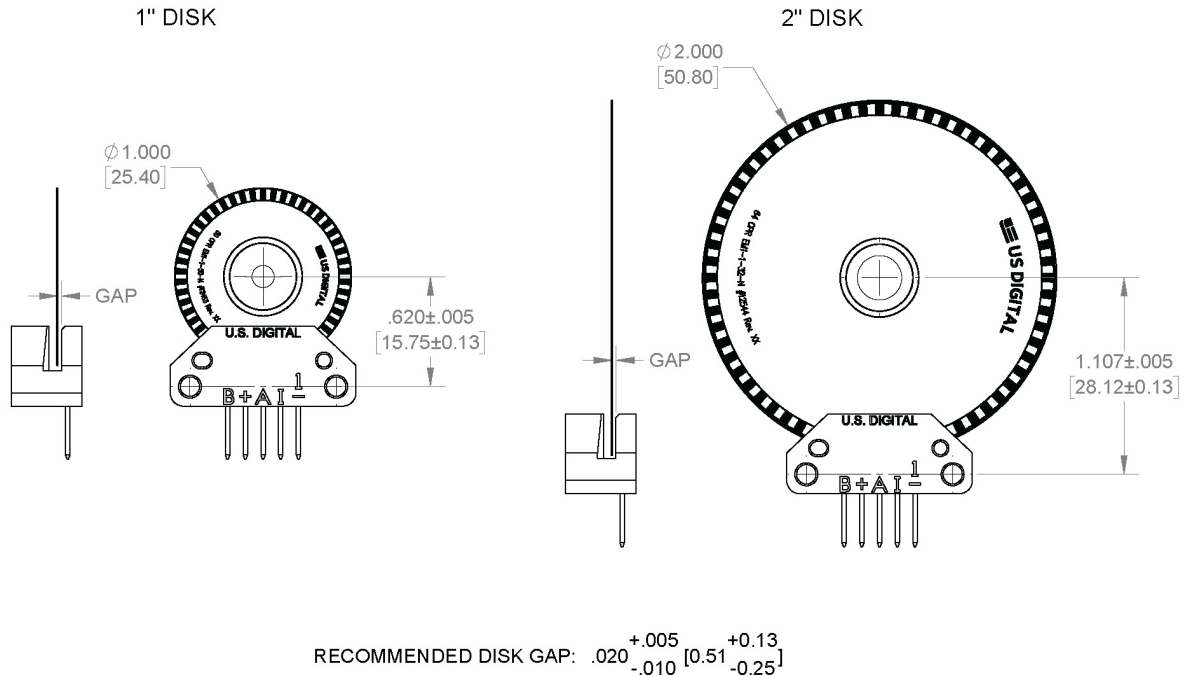
info@usdigital.com
www.usdigital.com

Local: 360.260.2468
Toll-free: 800.736.0194

UNITS: INCHES [MM]
METRIC SHOWN FOR REFERENCE ONLY

EM2 Transmissive Optical Encoder Module Disk Alignment

RELEASE DATE: 1/9/2024



US DIGITAL 1400 NE 136th Avenue
Vancouver, Washington 98684, USA

info@usdigital.com
www.usdigital.com

Local: 360.260.2468
Toll-free: 800.736.0194

UNITS: INCHES [MM]
METRIC SHOWN FOR REFERENCE ONLY

Specifications

COMPATIBLE 1" & 2" DISKS



EM2 Transmissive Optical Encoder Module

1" DISKS

CPR	MODULE Index	1" DISK Index
2000	EM2-1-2000-I	DISK-1-2000-*-IE
2048	EM2-1-2048-I	DISK-1-2048-*-IE
2500	EM2-1-2500-I	DISK-1-2500-*-IE
4000	EM2-1-4000-I	DISK-1-4000-*-IE
4096	EM2-1-4096-I	DISK-1-4096-*-IE
5000	EM2-1-5000-I	DISK-1-5000-*-IE
	*Represents the bore size	

2" DISKS

CPR	MODULE Index	2" DISK Index
3600	EM2-2-3600-I	DISK-2-3600-*-IE
4000	EM2-2-4000-I	DISK-2-4000-*-IE
4096	EM2-2-4096-I	DISK-2-4096-*-IE
5000	EM2-2-5000-I	DISK-2-5000-*-IE
7200	EM2-2-7200-I	DISK-2-7200-*-IE
8000	EM2-2-8000-I	DISK-2-8000-*-IE
8192	EM2-2-8192-I	DISK-2-8192-*-IE
10000	EM2-2-10000-I	DISK-2-10000-*-IE
	*Represents the bore size	



COMPATIBLE 1" & 2" HUBDISKS

1" HUBDISKS		
CPR	MODULE Index	1" HUBDISK Index
2000	EM2-1-2000-I	HUBDISK-1-2000-*-IE
2048	EM2-1-2048-I	HUBDISK-1-2048-*-IE
2500	EM2-1-2500-I	HUBDISK-1-2500-*-IE
4000	EM2-1-4000-I	HUBDISK-1-4000-*-IE
4096	EM2-1-4096-I	HUBDISK-1-4096-*-IE
5000	EM2-1-5000-I	HUBDISK-1-5000-*-IE
*Represents the bore size		
2" HUBDISKS		
CPR	MODULE Index	2" HUBDISK Index
3600	EM2-2-3600-I	HUBDISK-2-3600-*-IE
4000	EM2-2-4000-I	HUBDISK-2-4000-*-IE
4096	EM2-2-4096-I	HUBDISK-2-4096-*-IE
5000	EM2-2-5000-I	HUBDISK-2-5000-*-IE
7200	EM2-2-7200-I	HUBDISK-2-7200-*-IE
8000	EM2-2-8000-I	HUBDISK-2-8000-*-IE
8192	EM2-2-8192-I	HUBDISK-2-8192-*-IE
10000	EM2-2-10000-I	HUBDISK-2-10000-*-IE
*Represents the bore size		

COMPATIBLE LINEAR STRIPS

LPI	MODULE Non-Index	Linear Strip Non-Index	MODULE Index	Linear Strip Index
1000	EM2-0-1000-N	LIN-1000-*-N	EM2-0-1000-I	LIN-1000-*-#
2000	EM2-0-2000-N	LIN-2000-*-N	EM2-0-2000-I	LIN-2000-*-#
* Represents length of Linear Strip			* Represents length of Linear Strip # Represents location of Index	

ENVIRONMENTAL

Parameter	Value	Units
Operating Temperature	-25 to 100	C
Electrostatic Discharge, IEC 61000-4-2	± 4	kV
Vibration (10Hz to 2kHz, sinusoidal)	20	G
Shock (6 milliseconds, half-sine)	75	G

OPERATING CONDITIONS

PARAMETER	MIN.	MAX.	UNITS	NOTES
A/B Output Frequency	0	360	kHz	2000, 2048, 2500 CPR (1") 3600, 4000, 4096, 5000 CPR (2") 1000 CPI (Linear)
	0	720	kHz	4000, 4096, 5000 CPR (1") 7200, 8000, 8192, 10000 CPR (2") 2000 CPI (Linear)
Disk RPM	0	(21.6 x 10 ⁶) / CPR	RPM	2000, 2048, 2500 CPR (1") 3600, 4000, 4096, 5000 CPR (2")
	0	(43.2 x 10 ⁶) / CPR	RPM	4000, 4096, 5000 CPR (1") 7200, 8000, 8192, 10000 CPR (2")
Linear Strip Speed	0	360	in./sec.	1000, 2000 CPI (Linear)
Disk/Linear Strip Radial Position Tolerance	± .005		inch	

ELECTRICAL SPECIFICATIONS

- Specifications apply over the entire operating temperature/voltage range.
- Typical values are specified at $V_{CC} = 5.0V$ and $25^{\circ}C$.

PARAMETER	MIN.	TYP.	MAX.	UNITS	CONDITIONS
Supply Voltage	4.5	5.0	5.5	V	Ripple < 100 mVpp
Supply Current		72	85	mA	no load
Low-level Output		0.25	0.5	V V	$I_{OL} = 5\text{ mA}$ no load
High-level Output	2.0	3.5		V V	$I_{OH} = -5\text{ mA}$ no load
Output Current Per Channel	-5		5	mA	
Load Capacitance			100	pF	
Output Rise Time		50		nS	$\pm 5\text{ mA load}$
Output Fall Time		50		nS	$\pm 5\text{ mA load}$

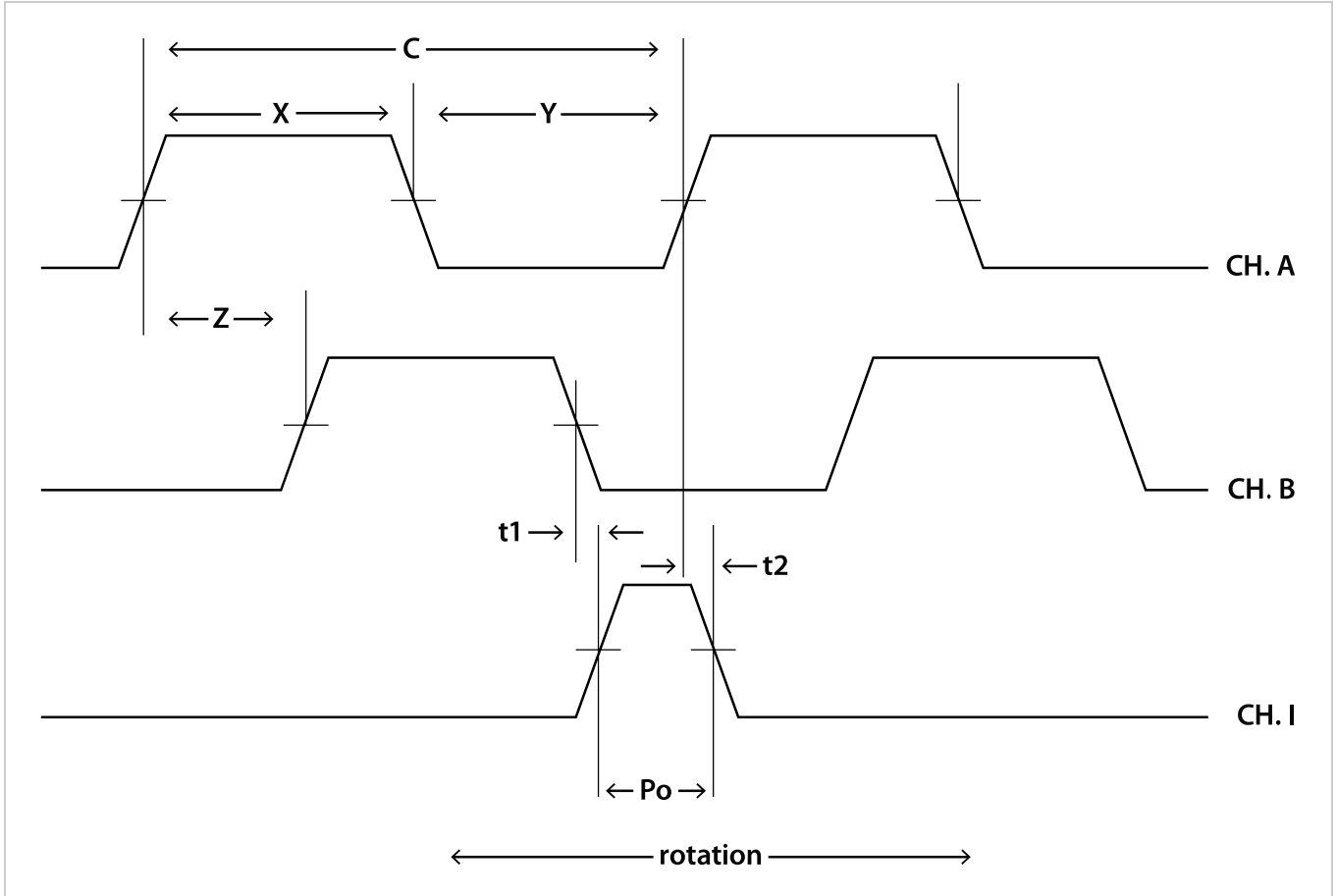
TIMING CHARACTERISTICS

ENCODING CHARACTERISTICS:

- Specifications apply over the entire operating temperature/voltage range.
- Values are for the worst error over full rotation.
- Refer to the timing diagram below.

PARAMETER	SYMBOL	MIN.	TYP.	MAX.	UNITS
Symmetry	X, Y	108	190	252	$^{\circ}e$ (https://www.usdigital.com/support/resources/glossary/electrical-degree/)
Quadrature	Z	45	90	135	$^{\circ}e$ (https://www.usdigital.com/support/resources/glossary/electrical-degree/)
Index Pulse Width	Po	40	90	135	$^{\circ}e$ (https://www.usdigital.com/support/resources/glossary/electrical-degree/)
Ch. I Rise After Ch. B or Ch. A Fall	t1		-40		ns
Ch. I Fall After Ch. B or Ch. A Rise	t2		25		ns

TIMING DIAGRAM:



CPI: The number of Cycles (C) of the A or B output Per Inch of linear strip movement.

CPR: The number of Cycles (C) of the A or B outputs Per Revolution.

Index (I): The index output goes high once per revolution, coincident with the low states of channels A and B, nominally 1/4 of one cycle (90 °e).

One Shaft Rotation: 360 mechanical degrees.

One Electrical Degree (°e): 1/360th of one cycle.

One Cycle: 360 electrical degrees (°e). Each cycle can be decoded into 1 or 4 states, referred to as X1 or X4 resolution multiplication.

PPR: The number of resolvable Positions Per Revolution of the encoder disk with x4 quadrature decoding.

Quadrature (Z): The phase lag or lead between channels A and B in electrical degrees, nominally 90 °e.

Symmetry: A measure of the relationship between (X) and (Y) in electrical degrees, nominally 180 °e.

INSTALLATION TORQUE

PARAMETER	TORQUE
Mounting Screws	3.5-4 in-lbs

PIN-OUTS

Pin	Description
1	Ground
2	Index
3	A channel
4	+5VDC power
5	B channel

Notes

- US Digital® warrants its products against defects in materials and workmanship for two years. See complete warranty (<https://www.usdigital.com/company/warranty>) for details.

Configuration Options

EM2	Type	Resolution	Index
	0 (<i>Linear Strip</i>)	1000 LPI	I (<i>Index</i>)
	1 (<i>1" Disk</i>)	2000 LPI	N (<i>Non-Index</i>)
	2 (<i>2" Disk</i>)	2000 CPR	
		2048 CPR	
		2500 CPR	
		4000 CPR	
		4096 CPR	
		5000 CPR	
		3600 CPR	
		7200 CPR	
		8000 CPR	
		8192 CPR	
		10000 CPR	

PLEASE NOTE: This chart is for informational use only. Certain product configuration combinations are not available. Visit the EM2 product page (<https://www.usdigital.com/products/EM2>) for pricing and additional information.